



Enhanced recovery of acetophenone and 1-phenylethanol from petrochemical effluent by highly porous starch-based hypercrosslinked polymers

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ABSTRACT

It is highly desirable to design and fabricate porous bio-based adsorbent for efficient and selective recovery of acetophenone (AP) and 1-phenylethanol (PE) from the effluent of propylene oxide and styrene monomer co-production (PO/SM) process. Herein, using cassava starch as the starting material, starch-graft-styrene hypercrosslinked polymers (SSHCPs) were designed and synthesized facily by starch-graft-styrene copolymerization and subsequent crosslinking via a Friedel-Crafts alkylation route. SSHCP-3 (knitted by 4,4'-bis-(chloromethyl)-1,1'-biphenyl) shows high BET surface area ($818 \text{ m}^2 \cdot \text{g}^{-1}$), large pore volume, and high thermal stability, making it a highly efficient adsorbent for removal of aromatic pollutants from water. SSHCP-3 showed high uptake for AP ($2.82 \text{ mmol} \cdot \text{g}^{-1}$) and selectivity (4.95) of AP over PE in binary-solute adsorption experiments. Spectral studies and theoretical calculation demonstrated that AP and PE uptake could occur at glucose units and benzene moiety via van der Waals interaction and π - π stacking interactions. SSHCP-3 also showed excellent adsorption efficiency of AP (93.6%) and PE (74.4%) in the wastewater discharged from PO/SM process. These results demonstrated that transforming styrene and starch to bio-based hypercrosslinked polymers provides an environmentally friendly and attractive strategy for recovery of AP and PE in practical application.

1. Introduction

Acetophenone (AP) and 1-phenylethanol (PE) are two important intermediates used in spices, flavors and pharmaceutical industries [1–3]. The propylene oxide and styrene monomer co-production technology (PO/SM process) produces a large amount of effluent containing mainly AP and PE. Unfortunately, it remains helpless to recover purified AP and PE from this concentrated wastewater on account of its complicated composition [4]. Instead, the petrochemical manufactures generally deal with the waste effluent with traditional incineration and biochemical treatment to meet the waste water discharge standard, resulting in a huge waste of resources. Therefore, it is reasonably practical to adopt a relatively effective way with low energy consumption to recover AP, PE and other components from the PO/SM waste effluent. Selective adsorption enrichment is considered to be an effective method to extract useful substances from industrial effluents. To improve the

efficiency of adsorption recovery of AP and PE, it is of critical importance to identify and develop adsorbents with high selectivity and large adsorption capacity.

To this end, the adsorption behaviors of AP and PE onto several low-cost adsorbents have been studied previously. For example, the kerolite clay was proved to have excellent selective adsorption behavior of AP over PE [5]. However, in spite of its distinct selectivity to AP over PE, the relatively low adsorption capacity (1.16 mmol g^{-1} for AP) limited its practical application. Recently, there is a growing interest in developing renewable bio-based alternative adsorbents with high adsorption capacity. Starch is one of the most attractive feedstock of biodegradable polymers owing to its biodegradability, abundance and comparatively low cost [6,7]. In contrast to the fascinating properties of starch, its disadvantages such as low specific surface area and less functional groups lead to low adsorption performance [8]. Therefore, chemical modification (e.g., grafting, crosslinking, etc.) of starch with various

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agents is supposed to be a feasible strategy to improve the textural and surface chemical properties, which endows the starch-based adsorbent with superior adsorption performance [9–11]. In previous studies, we constructed starch-based polymers by crosslinking starch with two aromatic diisocyanates, namely, bitolylene diisocyanate (TODI) and 4,4'-methylene diphenyl diisocyanate (MDI). Compared with other starch crosslinked polymers, the as-prepared aromatic-rich starch-based polymers exhibited greater preferential adsorption of AP over PE via π - π interaction [12,13]. However, for the above starch-based polymers, the further enhancement of selective adsorption was still hampered due to the unresolved intrinsic structural drawbacks of poor porosity and deficient specific surface area [14–16]. Hence, a feasible strategy to synthesize starch-containing polymers with high surface area is highly desirable for selective recovery of AP and PE from PO/SM effluent.

Hypercrosslinked polymers (HCPs) have received considerable attention in adsorption field due to the permanent microporous structure and high surface area [17–19]. Up to now, a large number of HCPs adsorbents have been prepared by post-crosslinking polystyrene-based precursors with external crosslinkers in the presence of Lewis acid catalyst and appropriate solvent [20–24]. Meanwhile, to fabricate bioplastic with enhanced mechanical strength, starch grafted with polystyrene (st-g-ps) was often prepared by suspension polymerization [25]. However, to the best of our knowledge, starch grafted with polystyrene was rarely used as the precursor for synthesizing bio-based HCPs adsorbents. Bearing this in mind, we aimed to graft polystyrene units onto starch chains, followed by introducing the external crosslinking agents to obtain starch-based HCPs with high surface area and permanent porosity. To manipulate both the suitable chemical composition and porosity of starch-based HCPs, we first synthesized starch-graft-styrene

copolymer and then hypercrosslinked it with three typical external crosslinkers, i.e., formaldehyde dimethyl acetal (FDA), dichloroxylene (DCX), 4,4'-bis-(chloromethyl)-1,1'-biphenyl (BCMBP), which contain methylene, xylene and biphenyl backbones, respectively. With the obtained polymers, the adsorption properties of AP and PE in aqueous phase were studied systematically, and then the selective extraction of AP/PE from PO/SM effluent was tentatively tested. Finally, the selective adsorption mechanism of AP over PE was investigated through theoretical calculation, providing a theoretical basis for designing the industrial bio-based adsorbents for the recovery of useful components from petrochemical process wastewater.

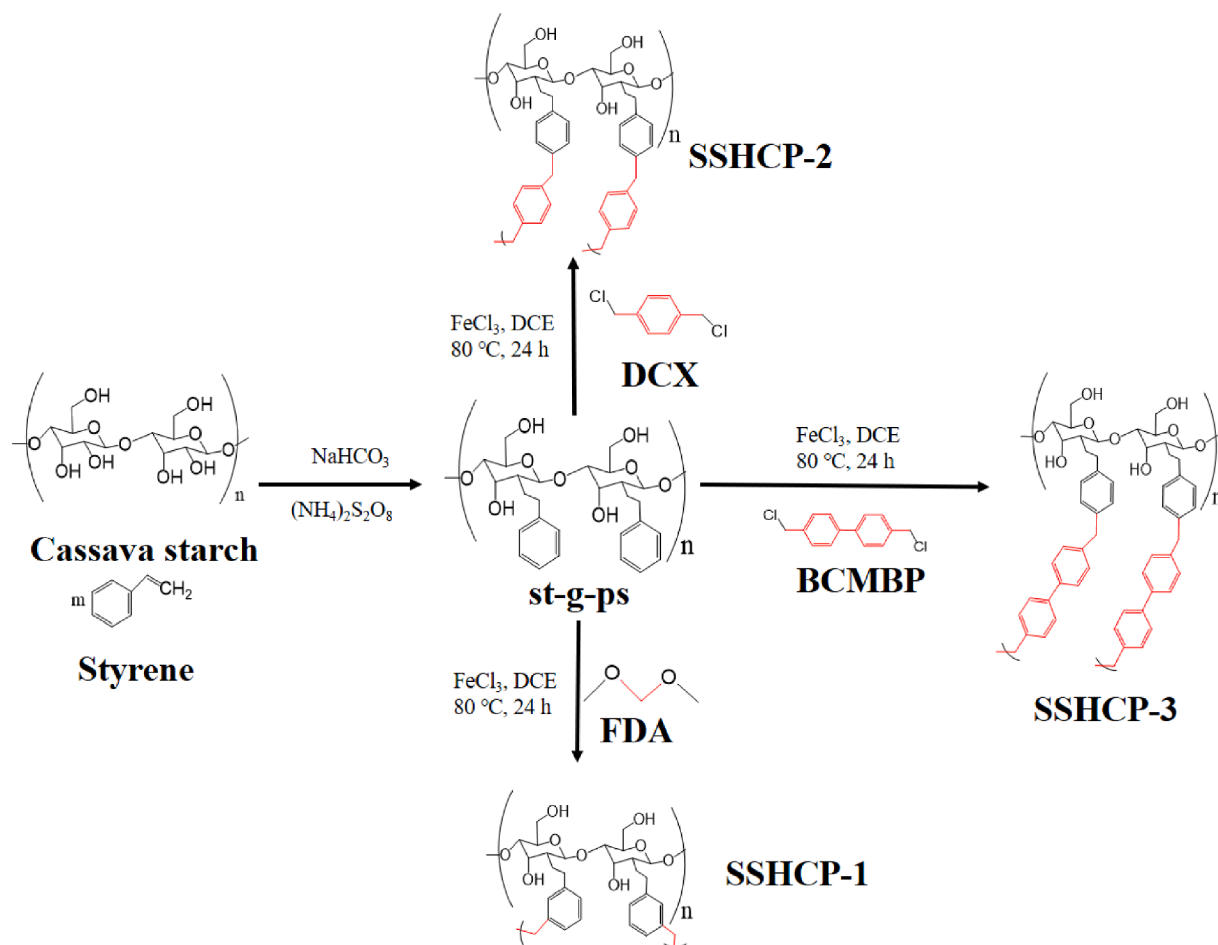
2. Materials and methods

2.1. Materials

Cassava starch was purchased from Guangxi State Farms Mingyang Biochemical Group. Styrene, FDA, DCX, BCMBP, anhydrous FeCl_3 , 1,2-dichloroethane (DCE) and other chemicals were analytical grade and obtained from Aladdin Shanghai Reagent Co., Ltd. The PO/SM effluent was obtained from Huizhou refinery branch of China National Offshore Oil Corp., CNOOC.

2.2. Synthesis of SSHCPs

The preparation routes to SSHCPs are shown in Scheme 1. Starch-graft-styrene copolymer was synthesized according to the article [26], and the graft percent was about 14%. A typical experimental procedure for SSHCP-1: a mixture of starch-graft-styrene copolymer (0.743 g) and



Scheme 1. Synthetic strategy of SSHCPs.

FDA (0.076 g, 1 mmol) was swelled overnight in 20 mL of DCE at room temperature, then FeCl_3 (0.649 g) was added and heated to 80 °C and stirred for 24 h under nitrogen atmosphere. After reaction, the brown precipitate was filtered off and washed with methanol until the filtrate was nearly colorless. Further purification of the polymers was carried out by Soxhlet extraction with methanol for 48 h. The polymers were dried in a vacuum for 24 h at 60 °C to give dark brown powder (yield: 75–80%). SSHCP-2 and SSHCP-3 were prepared with crosslinkers DCX (0.1750 g, 1 mmol) and BCMBP (0.2511 g, 1 mmol) following the same procedure as described for SSHCP-1.

2.3. General analysis

Fourier-transform infrared (FTIR) spectra were recorded using a Bruker TENSOR II spectrometer. Thermal degradation behavior of the samples was studied using Thermal Gravimetric Analyzer (TGA, Mettler Toledo) by heating the samples within the temperature range 30–800 °C at a heating rate of 10 °C·min⁻¹ under N₂ condition. The contact angles (CA) of the polymers were measured by Kruss DSA100 contact angle measuring instrument. X-ray diffraction (XRD) patterns were achieved on a X'Pert PRO diffractometer (PANalytica B.V., Holland) with Cu K α radiation under a voltage of 40 kV. The morphology of adsorbents was observed by scanning electron microscopy (SEM, Hitachi S-3400) and transmission electron microscopy (TEM, FEI Tecnai G2 F20, America). N₂ adsorption–desorption isotherms were measured on Micromeritics ASAP2460 instrument at -196 °C. All the samples were degassed under reduced pressure (<0.05 mbar) at 393 K for 24 h in the nitrogen adsorption analysis. Zeta potentials of adsorbent suspension in the pH range 1.0–13.0 were recorded on the NanoBrook Omni (Brookhaven, American). The specific surface areas were calculated using Brunauer-Emmett-Teller (BET) method and pore size distribution curves were then calculated by both NLDFT and BJH method, respectively. The corresponding surface elemental investigation of SSHCP-3 before and after adsorption were carried out by X-ray photoelectron spectrometer (XPS, ESCALAB-250XI, Thermo Fisher). The concentrations of AP, PE and other organics in various solutions were determined by gas chromatography (GC, Agilent 4890D) with naphthalene as internal standard [12].

2.4. Batch adsorption experiments

To investigate the adsorption behaviors onto SSHCPs, all the solutions were prepared by dissolving single or binary solutes of AP and PE in deionized water to the desired concentrations. Typically, adsorbent (0.1 g) was added into the initial solution (10 mL) and the mixture was shaken for a certain period of time. The solution was then filtered with 0.22 μm PTFE filter and the residual substrate concentrations were measured with gas chromatography (GC) [13]. For static kinetic experiments, adsorbent was added into 40 mmol L⁻¹ AP or PE solution, which were shaken for specific time periods. For single-solute adsorption isotherms, 10 mL of AP or PE solution with initial concentration from 20 to 60 mmol L⁻¹ were used and shaken at 25, 35 or 45 °C for 60 min. For studying the effect of different pH₀ values (2–12) on adsorption, the equimolar AP/PE solutions (both the concentrations of AP and PE were 40 mmol L⁻¹) were used and contact time was 60 min. For the recycle tests, adsorbent was added into the equimolar binary solution containing 40 mmol L⁻¹ of AP/PE, which were shaken at 25 °C for 60 min. After each use, the spent adsorbent was washed by ethyl acetate and dried overnight at 50 °C in the vacuum dryer. The amount of AP or PE adsorbed as well as selectivity was calculated based on mass balance. All these models and equations were summarized in [Supplementary Information](#). All sorption tests were performed in duplicate and the mean values with error bars were presented.

2.5. Computational methods

All calculations were performed using the Gaussian 16 program package. A starch glucose unit and a benzene moiety were used to simulate the adsorbent binding sites. The geometries and energy of AP, PE, glucose unit and benzene moiety were optimized by the density functional theory (DFT) method (M06-2X), as applied to study the π -stacking interaction on aromatic system. All calculations were performed at 6-31G* level and the vibrational frequencies were also calculated until there were no imaginary frequencies to ensure the stability of optimal structure [27]. The binding energy (BE) of AP or PE on glucose or benzene moiety were calculated as Eq.(1) [28]:

$$BE = E_{\text{com}} - (E_{\text{sub}} + E_{\text{ads}}) \quad (1)$$

where E_{com} , E_{ads} and E_{sub} stand for the total energies of complexes, adsorbent and substrate, respectively.

3. Results and discussion

[Scheme 1](#) shows the general synthetic routes for the three SSHCPs. Firstly, starch was grafted with styrene to introduce aromatic ring frameworks and then the Friedel-Crafts alkylation reaction was carried out using FDA, DCX and BCMBP as the external crosslinkers to prepare hypercrosslinked polymers. The resulting polymers were named as SSHCP-1, SSHCP-2 and SSHCP-3, respectively.

3.1. Physical characteristics and surface morphology

The comparisons of FTIR spectra of cassava starch and starch-based polymers are shown in [Fig. 1](#). The peak at about 3400 cm⁻¹ is attributed to the -OH stretching, and bands at 1000–1200 cm⁻¹ are associated with C-O stretching vibration of C-OH group and C-O-C group in starch anhydroglucose ring [29,30]. Compared with the FTIR spectra of starch, there narrows the peak at 3400 cm⁻¹ belonging to -OH of st-g-ps and SSHCPs, revealing -OH of starch participates in the graft polymerization. For st-g-ps and SSHCPs, feature bands at 1500–1600 cm⁻¹ are attributed to C = C vibration in the benzene skeleton [31], suggesting graft polymerization of polystyrene among starch chains. Meanwhile, a new peak appeared at 3020 cm⁻¹ belonging to Ar-H stretching [32]. The benzene ring mono-substituted characteristic peaks at 750 and 700 cm⁻¹ prove the abundant existence of polystyrene units in st-g-ps. However, the intensities of these two distinct characteristic peaks for SSHCPs become weaker, which may be due to the Friedel-Crafts alkylation of benzene ring on the polystyrene units and the content of mono-substituted polystyrene decreased in SSHCPs. The peak for SSHCP-1 at 850 cm⁻¹ indicates that there is *ortho*-substitution on the aromatic ring. The peak for SSHCP-2 and SSHCP-3 at 800 cm⁻¹ manifests that there are counterpoint substitutions that cross-linking reaction on the *para*-position of polystyrene with DCX or BCMBP.

Thermal stability of the obtained SSHCPs was studied by thermogravimetric analysis (TGA) ([Fig. S1](#)) in N₂ atmosphere. The weight loss of native starch was more than that of hypercrosslinked polymers at the temperature above 800 °C, suggesting the high thermal stability of the hypercrosslinked network. It is worthy to note that the thermal stabilities are in the order of SSHCP-1 < SSHCP-2 < SSHCP-3, which could be attributed to increase of aromatic rings in crosslinking units of SSHCPs. Such a high thermal stability could be beneficial for the regeneration of adsorbents in practical applications.

The morphology nature of SSHCPs was observed by SEM and TEM. As illustrated in [Fig. 2](#), the smoother surface of SSHCP-1 is ascribed to embeddedness of flexible methylene groups than original cassava starch. Nevertheless, the surface of SSHCP-2 and SSHCP-3 are rougher due to the successful loading of rigid aromatic ring. The XRD patterns ([Fig. S2](#)) showed the crystallized cassava starch was grafted with styrene and then hypercrosslinked to amorphous polymers. It can be observed from the

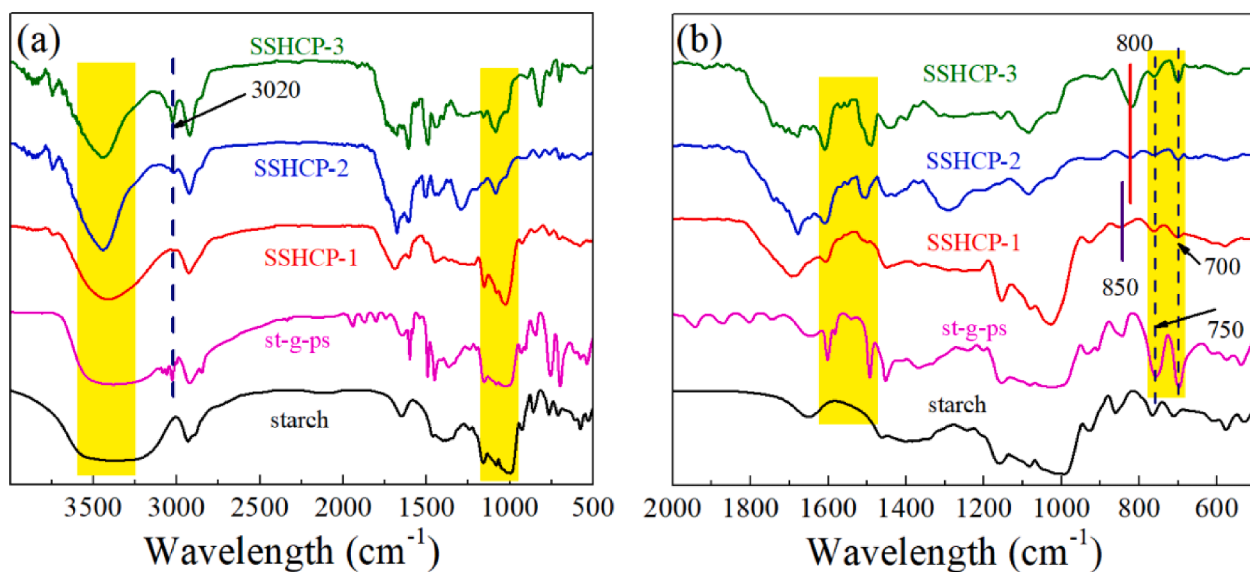


Fig. 1. FTIR spectra of starch and starch-based polymers (a) full view and (b) partially enlarged view.

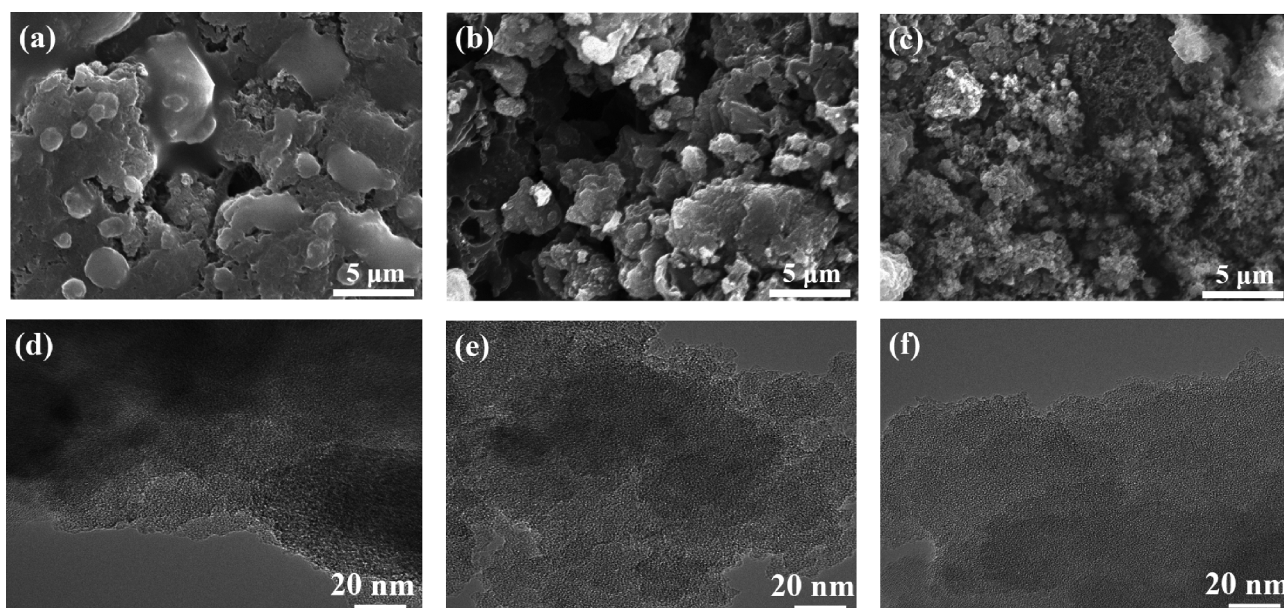


Fig. 2. The SEM images of (a) SSHCP-1, (b) SSHCP-2, (c) SSHCP-3; TEM images of (d) SSHCP-1, (e) SSHCP-2 and (f) SSHCP-3.

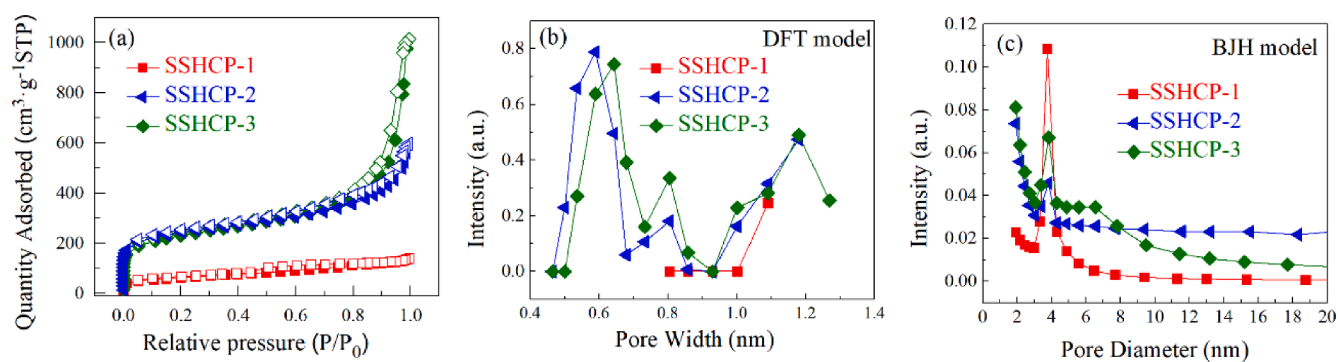


Fig. 3. (a) Nitrogen adsorption-desorption isotherms for the SSHCPs (the adsorption branch is labeled with filled symbols and the desorption branch is labeled with open symbols), (b) Pore size distribution curves calculated by NLDFT and (c) Pore size distribution curves calculated by BJH.

TEM images that numerous pores are distributed in the framework of these polymers. In addition, by roughly comparing the TEM images of SSHCPs, the pore density and distribution of SSHCP-3 are more uniform than that of SSHCP-1 and SSHCP-2, indicative of the highest porosity of SSHCP-3 among these polymers.

Direct characterization of the porous structures was carried out by N₂ adsorption–desorption measurements. As shown in Fig. 3(a), the Type I isotherms confirm that SSHCPs have predominantly microporous structure [33]. Furthermore, in the middle and high pressure phase, the hysteresis loop appeared in the isothermal curves, indicating the existence of mesoporous hole in the sample. Brunauer-Emmett-Teller (BET) surface areas were estimated to be 222, 791 and 818 m² g⁻¹ for SSHCP-1, SSHCP-2 and SSHCP-3, respectively (Table 1). Notably, the more the number of aromatic rings of the crosslinker monomer, the higher the specific surface area of SSHCPs, which should be resulted from that the aromatic ring in DCX and BCMBP shows comparatively more rigid nature than the –CH₂– in FDA [34]. Fig. 3(b) and (c) show that all the samples exhibited abundant microporous structures (pore diameters < 2.0 nm) with a spot of mesopore peaked at around 3.9 nm. Clearly, the aromatic ring structure is rigid enough to form microporous polymers with a high BET surface area, which is in line with the pore size distribution and TEM images. SSHCP-3 possesses a high surface area that could undoubtedly endow the material with high adsorption capacity.

3.2. Batch adsorption studies

3.2.1. Adsorbent screening

The selective adsorption performance tests were conducted in binary-solute (AP and PE) solution and the results are presented in Table S1. Compared with native cassava starch, both adsorption capacities and selectivity of SSHCPs increased significantly, which may be attributed to its higher specific area and functional groups. These results were also consistent with the SEM and N₂ adsorption–desorption results. Furthermore, the contact angle (CA) of SSHCP-3 was measured to be 130.2°, much higher than SSHCP-1 of 87.4° (Fig. S3). As control experiment, PS-BCMBP prepared by directly hypercrosslinking styrene with BCMBP and the CA in video S1 showed that it is super hydrophobic material. Although the hydrophobicity of materials is beneficial to adsorb hydrophobic organic pollutants in aqueous solution, PS-BCMBP is too hydrophobic to disperse well in aqueous solution, which is not conducive to the practical water treatment. Given the dispersion of adsorbent in water and adsorption performance for AP and PE, SSHCP-3 was adopted as a representative for further adsorption exploration.

3.2.2. The single-solute adsorption kinetic studies

Results from the single-solute adsorption kinetic study (Fig. 4(a)) demonstrated the apparent adsorption equilibrium was achieved within 10 min. In particular, the curves show that the adsorption capacity of SSHCP-3 increased rapidly in the first 5 min. It may be because that the

Table 1
Surface areas and pore volumes of starch and SSHCPs.

Sample	S _{BET} (m ² g ⁻¹)	S _{micro} (m ² g ⁻¹)	S _{meso} (m ² g ⁻¹)	V _{total} (cm ³ g ⁻¹)	V _{micro} (cm ³ g ⁻¹)	V _{meso} (cm ³ g ⁻¹)
SSHCP-1	222	155	67	0.21	0.08	0.13
SSHCP-2	791	400	391	1.57	0.31	1.26
SSHCP-3	818	324	494	0.92	0.33	0.59

S_{BET}: the BET surface area.

S_{micro} and S_{meso}: the surface areas of microporous and mesoporous structure, respectively.

V_{total} and V_{micro}: represent the total volume and microporous volume, respectively.

adsorption sites were abundantly available at the beginning of adsorption process. In order to confirm the adsorption mechanism, data were fitted to the well-known pseudo first-order (PFO) and pseudo second-order (PSO) kinetic models, using non-linear regression [35], and the kinetic parameters and correlation coefficients are shown in Table S3. It can be seen, PSO kinetic model is appropriate to describe the adsorption process toward AP and PE because of its higher correlation coefficients ($R^2 > 0.999$). The q_e values of AP and PE (3.23 and 2.37 mmol·g⁻¹) are higher than that of in our previous research (2.39 and 0.99 mmol·g⁻¹ for AP and PE, respectively) [36], which would be ascribed to the high surface area of SSHCP-3.

3.2.3. The single-solute adsorption isotherms

AP and PE adsorption isotherms were recorded to gain further insight into the affinity and saturation values of SSHCP-3 (Fig. 4(b)). The adsorption capacities of both solutes increase significantly as initial concentrations increase gradually at all temperatures. It is observed q_e decreases as the temperature increases, indicating low contact temperature is favorable for adsorption. To further investigate the adsorption isotherms, single-solute isotherms experimental data were fitted to Langmuir and Freundlich adsorption models [37], and the parameters of isotherm models are listed in Table S3. Freundlich adsorption model was more suitable to describe the adsorption behavior because of the higher regression coefficient ($R^2 > 0.99$). The results suggested that the adsorption of AP and PE both take place heterogeneously on the surface of onto SSHCP-3 by multilayer adsorption.

3.2.4. The single-solute adsorption thermodynamics

Thermodynamics parameters, which can be calculated from the temperature dependent adsorption isotherms, are evaluated to investigate the nature of adsorption process. Basically, the adsorption enthalpy ΔH^0 for physical adsorption was in the range of 2.1–20.9 kJ mol⁻¹, and 80–200 kJ mol⁻¹ for chemisorption [38]. As can be seen in Table 2, negative ΔH^0 (its absolute value < 10 kJ·mol⁻¹) and ΔS^0 revealed an exothermic physical adsorption process with decrease of freedom degree at the liquid–solid interface. Compared with PE, the more negative ΔG^0 was observed for AP uptake onto SSHCP-3, indicating that the preferential adsorption of AP over PE [39].

3.2.5. Competitive adsorption of AP and PE

It is necessary to study the competitive adsorption behaviors of AP and PE because AP and PE coexist as main components in PO/SM effluent. Fig. 5(a) shows the different concentrations of AP/PE (equal molar ratio) binary-solute solutions influenced on adsorption performance of SSHCP-3 at 25 °C. The adsorption capacities are in the order of AP > PE, indicating the preferential adsorption for AP over PE. Moreover, the AP/PE selectivity (about 4.95) was almost unchanged with the increase of the concentrations of solutes. The concentration-independent selectivity may be attributed to the high adsorption capacities of SSHCP-3 for AP and PE. The uptake also increases with the increase of concentration, which should be owed to the large adsorption capacity caused by the high specific surface area of the material.

3.2.6. pH influence on competitive adsorption of AP/PE and SSHCP-3 regeneration experiments

In order to evaluate the stability of SSHCP-3 in the adsorption process, the effect of solution pH₀ on competitive adsorption and regeneration tests were carried out. The influence of pH₀ on the adsorption of AP and PE was explored by adding a few drops of HCl or NaOH solution to obtain solutions with pH values ranging from 2 to 12. Although zeta potential of SSHCP-3 under different pH values (1.0–13.0) showed that the pH of solution can change the surface charge of the adsorbent (Fig. S4), both the adsorption capacities and selectivity kept stable in the investigated pH₀ range (Fig. 5(b)), suggesting that electrostatic interaction played a negligible role in the adsorption. The high adsorption capacity over a wide pH range is crucial in practical applications.

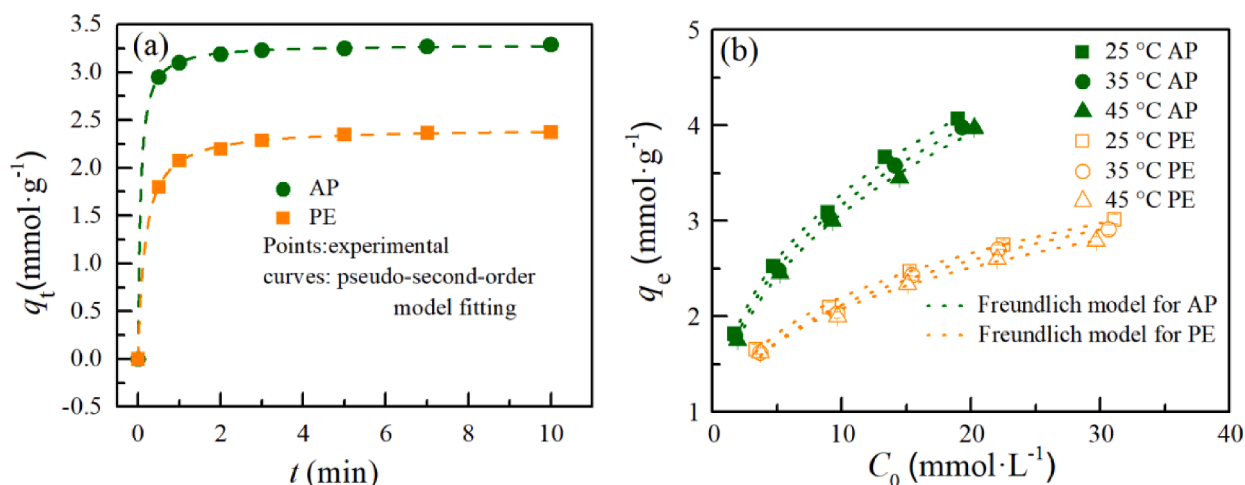


Fig. 4. (a) Effect of time on the adsorption of AP and PE onto SSHCP-3 in aqueous solution at 25 °C and (b) adsorption isotherms of AP and PE on SSHCP-3 fitted with the Freundlich isotherm model by non-linear regression at 25, 35 and 45 °C.

Table 2

Thermodynamic parameters of AP and PE adsorption by SSHCP-3.

Adsorbate	K			ΔG^0 (kJ·mol ⁻¹)			ΔH^0 (kJ·mol ⁻¹)	ΔS^0 (J·mol ⁻¹ ·K ⁻¹)
	25 °C	35 °C	45 °C	25 °C	35 °C	45 °C		
AP	3.56	3.27	2.78	-3.15	-3.03	-2.71	-9.74	-21.98
PE	3.16	2.42	2.31	-2.85	-2.27	-2.22	-12.36	-32.16

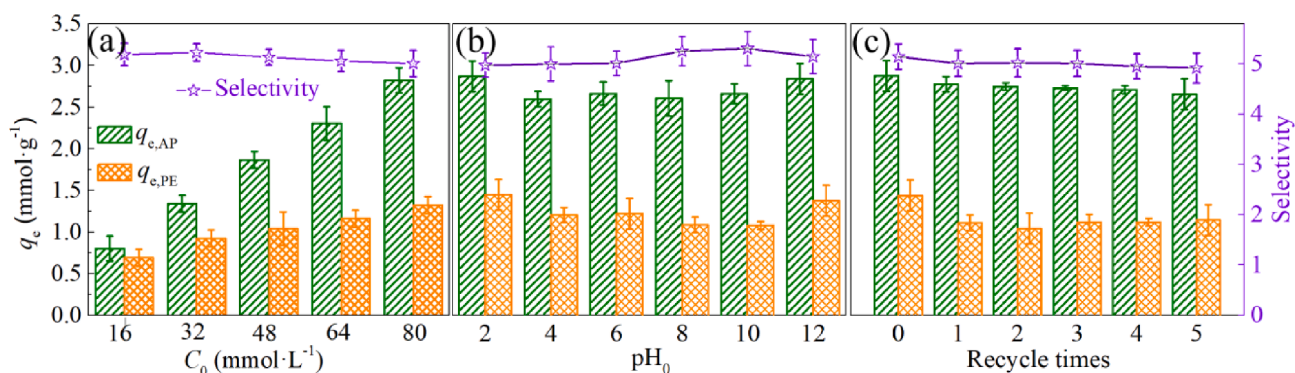


Fig. 5. (a) The performance of SSHCP-3 in binary-solute adsorption experiments, (b) Influence of pH_0 value on the binary-solute competition adsorption and (c) Regeneration experiments for SSHCP-3.

Furthermore, the adsorption capacity and selectivity had no obvious reduction during five recycle adsorption-desorption runs (Fig. 5(c)), proving high recyclability and stability of SSHCP-3 for AP and PE adsorption-desorption process.

In addition, compared with other adsorbents for AP and PE, SSHCP-3 show higher adsorption capacities while maintaining good selectivity for AP. Besides, the uptake of AP and PE can be attained equilibrium within 5 min (Fig. 6(a)). Compared with aryl-rich starch polymers (starch-TODI and starch-MDI), SSHCP-3 can enhance adsorption capacity of aryl-rich starch-based materials because SSHCP-3 with high surface area provides more available adsorption sites for AP and PE and mesoporous structure of SSHCP-3 is highly desired to obtain fast adsorption kinetics. For cellulose-TTI and β -CD-TDIT, the high adsorption selectivity should be due to abundant aromatic groups and hydrophobic cavity of β -CD, respectively [12,13,36,40]. Therefore, starch-based adsorbent with high specific surface and pore structure formed by hypercrosslinking could facilitate the adsorption of AP and PE.

3.2.7. PO/SM effluent treatment

Normally, an actual chemical waste water does not contain equal amounts of AP and PE as applied for most experiments described above, and it contains a high concentration of PE, a relatively low concentration of AP and other organic residues. Therefore, the practical applicability of SSHCP-3 was tested in PO/SM effluent. Comparing the GC graph and the concentration-change column graphics of PO/SM effluent before and after adsorption (Fig. 6(b, c), Table S4), it can be seen that the concentration changes of AP and PE were notable, and their adsorption efficiencies were 93.6% and 74.4%, respectively. Besides, the adsorption efficiency order of other coexist organics is phenol (38.1%) > benzyl alcohol (47.3%) > toluene (5.2%) > ethylbenzene (3.5%). Moreover, the selectivity of SSHCP-3 for AP over PE is 5.17. Therefore, SSHCP-3 still maintains a high adsorption capacity and selectivity for AP in the PO/SM effluent treatment.

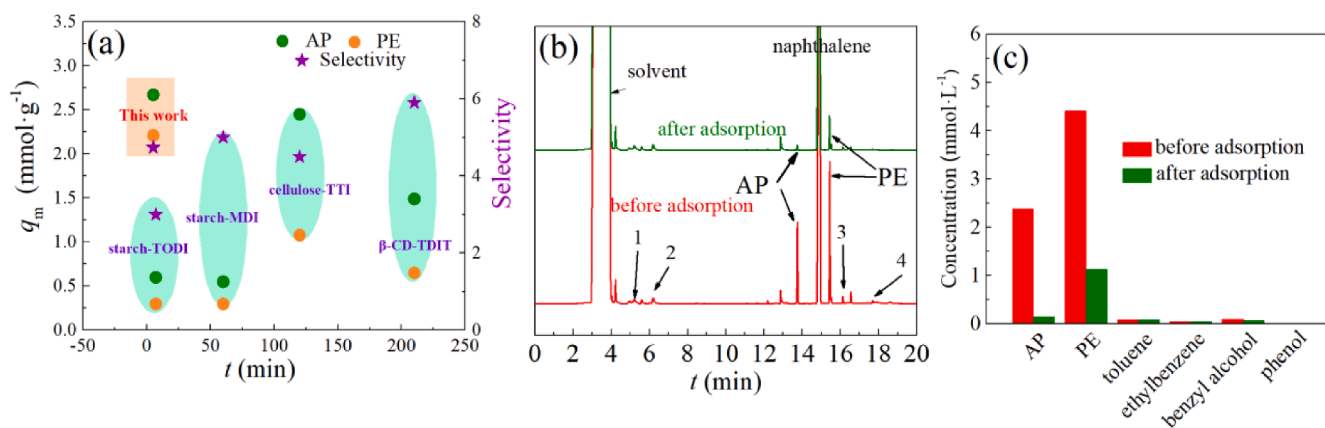


Fig. 6. (a) Comparison of AP and PE adsorption capacity (q_m) and equilibrium adsorption time for our prepared adsorbents with other reported adsorbents, starch-TODI [13], starch-MDI [12], cellulose-TTI [40] and β -CD-TDIT [36], (b) Comparison of GC chromatograms of PO/SM effluent before and after adsorption and (c) Histogram of typical organics concentrations before and after adsorption.

3.3. Suggested adsorption mechanism

For better clarifying the adsorption mechanism, XPS and FTIR of SSHCP-3 after adsorption of AP or PE are performed. The FTIR spectra in Fig. S5 revealed that the spectra blue-shift of C = C stretching from 1443 cm⁻¹ to 1455 cm⁻¹ after AP and PE were adsorbed, indicating that AP and PE adsorption could occur at benzene moieties [37]. The high-resolution XPS spectra (C1s) of SSHCP-3 before and after AP or PE adsorption are shown in Fig. 7. The C1s spectra show peaks at the binding energies of 284.3, 285.5 and 290.6 eV, which are assigned to C-C/C-H/C = C, C-OH/C-O-C and π - π^* shake-up satellite peaks, respectively [41–43]. The increase of O1s relative content from 9.81% to 14.94% and 14.15% indicates that SSHCP successfully adsorbed AP and PE (Fig. S6). For SSHCP-3/AP, a new small peak at 287.0 eV can be ascribed to the carbon atoms in carbonyl of AP molecule (C = O) appears, which is absent in the C1s spectra of SSHCP-3 and SSHCP-3/PE. For SSHCP-3/PE, the chemical shift of C-O moved to higher energy (285.5 $\rightarrow$ 286.1 eV) because of the increase of hydroxyl group after adsorption of PE. From the binding energies of C1s (C-C/C-H/C = C) and O1s spectrum of SSHCP-3 remained virtually unchanged after adsorption, indicating AP and PE have been adsorbed onto SSHCP-3 through physical adsorption involving formation of noncovalent interactions [44].

To further demonstrate the intrinsic driving force for AP or PE adsorption by SSHCP-3, we performed DFT studies to analyze the interaction between the repeating unit (glucose unit and benzene moiety) as fragments of SSHCP-3 and AP or PE. By calculating the binding energies of SSHCP-3 with AP or PE, typical π - π interaction can be observed between benzene rings from SSHCP-3 (Fig. 8(a) and (b)).

Compared with PE, the π - π stacking interaction between benzene ring and AP is stronger (indicated by the higher adsorption energy), which may be caused by that AP possess the electron-withdrawing group (C = O) that reduces the electron repulsion force of π -electron and increases the π - π interaction [45]. By comparing the length and angle of hydrogen bond in the optimized configuration, it was found PE has a stronger hydrogen bond with glucose than AP (Fig. 8(c)). However, glucose has higher affinity for AP as can be seen from the higher binding energy to AP, possibly because there are other stronger van der Waals forces between AP and glucose unit. Therefore, on the whole, both benzene ring unit and glucose unit of SSHCP-3 is more attractive to AP than PE, which agrees with the experimental results. Moreover, it can be seen that high specific surface area and abundant pore structure are conducive to the increase of adsorption capacity through the comparison of different starch-based adsorbents above. Therefore, van der Waals force, hole size effect, hydrogen bond and π - π stacking are involved in the adsorption process of AP and PE on SSHCP-3 (Fig. 9).

4. Conclusion

In summary, a series of hypercrosslinked polymers have been successfully synthesized via FeCl₃-mediated Friedel-Crafts alkylation polymerization of starch-graft-styrene copolymer using FDA, DCX, and BCMBP as external crosslinkers. Among them, SSHCP-3 showed high BET surface area (818 m²·g⁻¹), large pore volume, and high thermal stability. It was demonstrated that SSHCP-3 displays a promising combination of high AP adsorption capacity, desirable AP selectivity over PE, and fast sorption kinetics. Furthermore, it not only retains high adsorption efficiency (93.6% for AP, 74.4% for PE) in PO/SM effluent

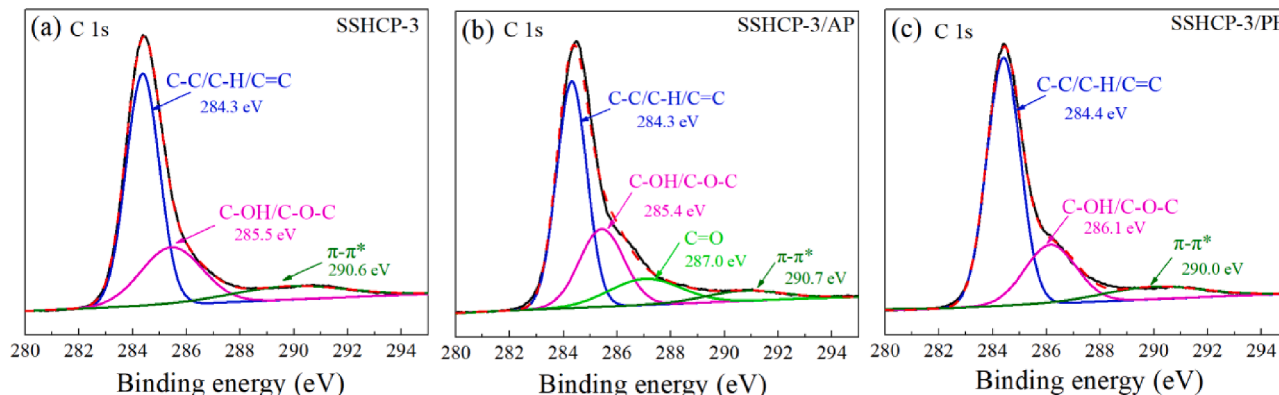


Fig. 7. XPS C1s of (a) pristine SSHCP-3, (b) SSHCP-3/AP and (c) SSHCP-3/PE.

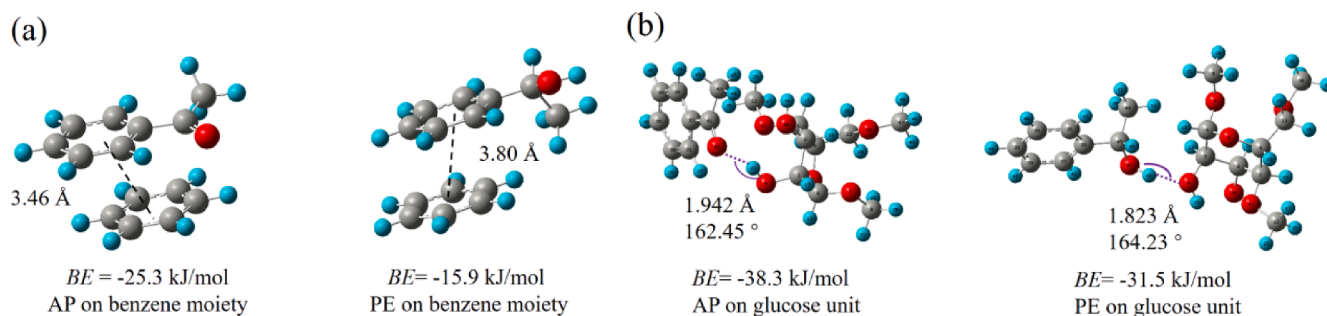


Fig. 8. Optimized geometrical structures of the adsorption of (a) AP or PE on benzene moiety and (b) starch glucose unit. Carbon: gray, Oxygen: red, Hydrogen: aqua green.

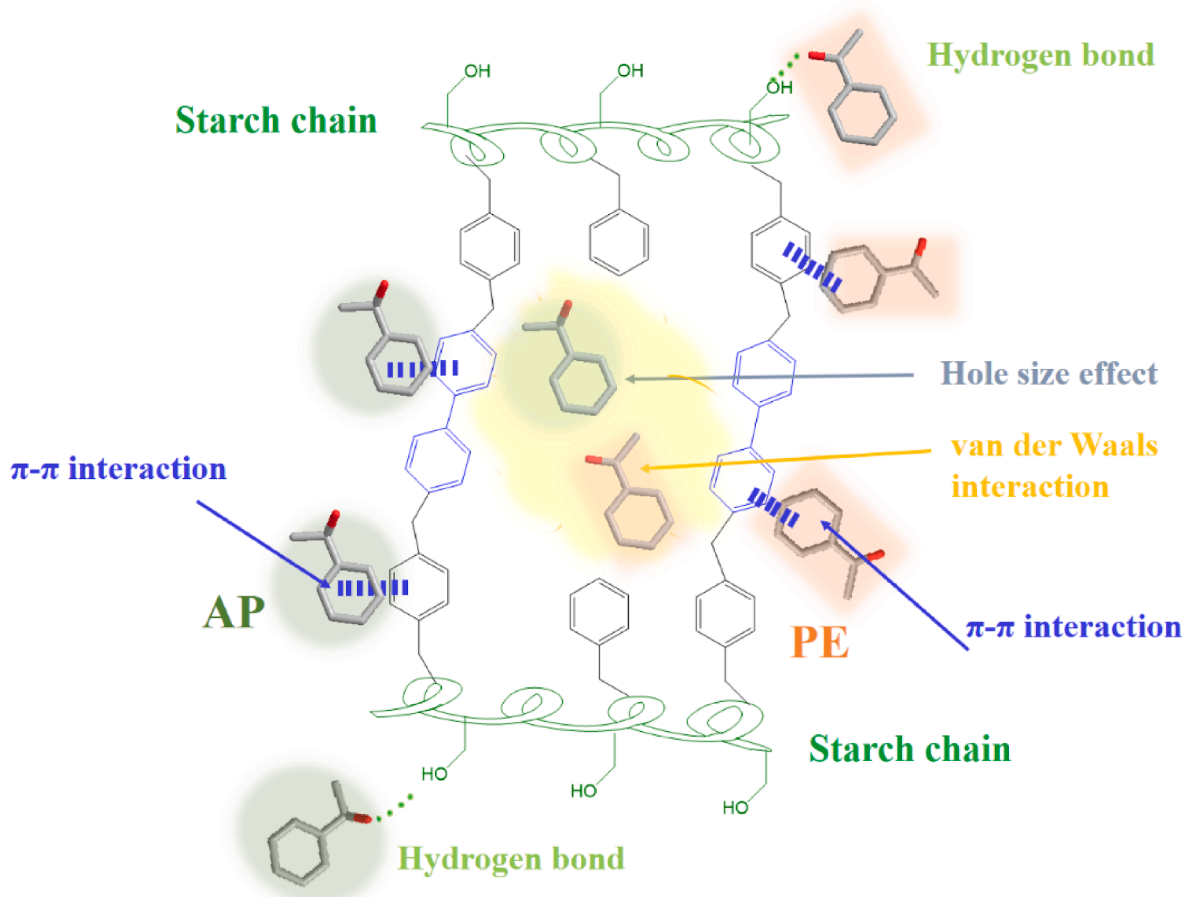


Fig. 9. The proposed adsorption mechanism of AP and PE onto SSHCP-3.

adsorption process but also has outstanding stability under repeated adsorption–desorption cycling. The theoretical calculations proved that π - π stacking interactions, van der Waals and hydrogen bond were involved the adsorption-separation process. Therefore, SSHCP-3 revealed the great potential as an attractive adsorbent for efficient and selective recovery of AP and PE in PO/SM effluent.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2021.129351>.

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